

Review of Existing Psychological Assessment Tools and Supporting Research

Project Title: InnerBloom- Mental Wellbeing Management System

Prepared By: Sumaiya Anjum

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1. Standardized Psychological Assessment Tools

PHQ-9 (Patient Health Questionnaire-9)

The PHQ-9 is one of the most widely used depression screening tools. It contains nine questions based on the DSM criteria for depression and has been validated across different populations with high sensitivity and specificity.

How it supports our system

- ❖ Used as the depression assessment questionnaire.
- ❖ Automatically calculates depression severity.
- ❖ Helps users recognize possible symptoms early.
- ❖ Allows psychologists to review standardized results before consultation.

GAD-7 (Generalized Anxiety Disorder-7)

The GAD-7 is a validated questionnaire consisting of seven questions used to measure anxiety severity. It is commonly used in hospitals, clinics, and online mental health services.

How it supports our system

- ❖ Used for anxiety screening.
- ❖ Provides automatic severity classification.
- ❖ Helps psychologists prioritize users needing immediate support.
- ❖ Enables users to monitor anxiety levels over time.

2. Review of Related Research

Research Paper 1

Kroenke, K., Spitzer, R. L., & Williams, J. B. W. (2001). *The PHQ-9: Validity of a Brief Depression Severity Measure*.

This study demonstrated that the PHQ-9 is a reliable and valid instrument for detecting depression in primary healthcare settings. The questionnaire showed high accuracy and has since become one of the most commonly used depression screening tools worldwide.

Relevance to our system

- ❖ Confirms that using the PHQ-9 provides scientifically reliable depression screening.
- ❖ Supports the system's automatic depression score calculation.
- ❖ Improves credibility of the mental health assessment feature.

Research Paper 2

Spitzer, R. L., Kroenke, K., Williams, J. B. W., & Löwe, B. (2006). *A Brief Measure for Assessing Generalized Anxiety Disorder (GAD-7)*.

This research validated the GAD-7 questionnaire and showed that it has excellent reliability and accuracy for identifying anxiety disorders.

Relevance to our system

- ❖ Justifies the inclusion of GAD-7 for anxiety assessment.
- ❖ Supports automated anxiety severity reports.
- ❖ Ensures that anxiety screening is based on internationally accepted research.

3. How These Tools Improve Our System

Existing Tools / Research0	Contribution to Our System
PHQ-9	Depression screening and severity calculation
GAD-7	Anxiety screening and severity calculation
PHQ-9 Validation Research	Confirms reliability and scientific accuracy of depression assessment
GAD-7 Validation Research	Confirms reliability and scientific accuracy of anxiety assessment
Standardized Scoring Methods	Enables automatic result generation and assists psychologists during consultation